TimerStudio

User Guide

v2026.05.13.0002

5 Timer Modes • CalDAV + ICS • Pomodoro • Email & Webhook Alerts

Windows 10/11 • .NET 10 • WPF/MVVM

1. Overview

TimerStudio is a multi-timer productivity application supporting five independent timer modes: Countdown, Stopwatch, DateTime, Pomodoro, and Interval. Each timer is a self-contained card with its own alert sounds, overlay style, recurrence, category, and optional email or webhook notification.

Calendar integration (CalDAV and ICS URL feeds) pulls upcoming events into a timeline view. Email alerts support four delivery methods including SMTP (with DPAPI-encrypted password storage), Outlook COM automation, and OAuth2.

2. System Requirements

Component	Requirement
OS	Windows 10 v1903 or Windows 11
Runtime	.NET 10 Desktop Runtime (x64)
RAM	64 MB minimum
Network	Required for CalDAV/ICS sync and OAuth2/SMTP alerts
Outlook	Outlook COM alerts require Microsoft 365 Outlook desktop installed
Permissions	Standard user — no elevation required; DPAPI for password storage

3. Installation

Run the TimerStudio installer. Settings in %APPDATA%\TimerStudio\settings.json. Timers persisted in %APPDATA%\TimerStudio\timers.json. SMTP credentials stored via Windows DPAPI (not plain-text).

4. Interface Tour

Zone	Purpose
1 Identity/Scale	App name, version, DPI scale (100/125/150/200%)
2 NavRow	Timers | Calendar | Stats | Settings | Themes | About
3 Toggle/Preview	Global pause-all toggle + timer count summary
4 Main	Timer card list; expand a card to reveal its settings panel
5 Presets 1–5 / Save	Preset slots store timer layout + global settings; resize grip bottom-right

5. Timer Modes

Mode	Description	Typical Use
Countdown	Counts from a set duration to 00:00; fires alert at zero	Meeting timers, cooking, presentations

Stopwatch	Counts up from zero; supports lap times	Exercise, benchmarking, recording durations
DateTime	Counts down to (or up from) a specific date/time	Event countdowns, anniversary trackers
Pomodoro	Alternates Work / Short Break / Long Break phases	Focus sessions, study blocks
Interval	Repeats a work period + rest period indefinitely	HIIT, stretching reminders, regular check-ins

6. Creating & Configuring Timers

Press Ctrl+N or click + Add Timer. Select a mode and configure in the settings panel:

Setting	Description
Label	Display name on the timer card (max 40 chars)
Duration / Target	Mode-specific: duration for Countdown/Pomodoro/Interval; target date for DateTime
AlertSound	Beep / Chime / Alarm / Bell / Ping / Silent / Custom (CustomSoundPath)
AlertStyle	Lens (floating overlay) / Fullscreen / Both
AlertVolume	0–100%
LensPosition	9 positions: TopLeft/Top/TopRight/Left/Centre/Right/BottomLeft/Bottom/BottomRight
AlertRepeatLimit	How many times the alert fires before auto-dismiss (0 = until dismissed)
EscalateAfterSeconds	If unacknowledged, escalate to Fullscreen after N seconds
SnoozeDurationMinutes	Snooze button duration (default 5 min)
RecurrenceMode	None/Daily/Weekly/Monthly — auto-restarts the timer on completion
CardTint	Accent colour for the timer card background (ARGB)
ShowInOverlay	Show this timer in the overlay window (see §13)
Category	Assign to a category for grouping and bulk control
WebhookUrl	HTTP POST on alert (see §12)
Notes	Free-text notes displayed on card expansion

7. Pomodoro Mode

Pomodoro timers cycle through phases automatically:

Phase	Default Duration	Configurable Setting
Work	25 minutes	PomodoroWorkMinutes (1–120)
Short Break	5 minutes	PomodoroBreakMinutes (1–60)
Long Break	15 minutes	PomodoroLongBreakMinutes (1–120)

Long Break After	Every 4 work sessions	PomodoroLongBreakAfter (2–8)

The current phase (Work / Break / Long Break — PomodoroPhase enum) is shown on the card with a coloured indicator. Each phase transition triggers an alert. Skip a phase early with the Skip button on the card.

8. Interval Mode

Interval mode repeats a work period followed by a rest period indefinitely until manually stopped.

Field	Description
Work Duration	Active period length (hh:mm:ss)
Rest Duration	Rest/recovery period length
Repeat Count	Maximum cycles before auto-stop (0 = infinite)
Alert on Work Start	Fire alert when work phase begins
Alert on Rest Start	Fire alert when rest phase begins

9. Alert System

9.1 Alert Styles

Style	Description
Lens	Small floating overlay at LensPosition; click to dismiss or snooze
Fullscreen	Semi-transparent fullscreen overlay with dismiss/snooze; blocks interaction
Both	Lens appears first; escalates to Fullscreen after EscalateAfterSeconds

9.2 Alert Sounds

Built-in sounds: Beep, Chime, Alarm, Bell, Ping, Silent. Set CustomSoundPath to any .wav or .mp3 file for a custom sound. AlertVolume (0–100%) is independent of system volume.

9.3 Snooze

Snooze button appears in the Lens and Fullscreen overlays. SnoozeDurationMinutes (default 5) delays the next alert. Snooze is available up to AlertRepeatLimit times.

10. Calendar Integration

10.1 CalDAV

Connect a CalDAV server in Settings → Calendar → CalDAV. Enter the server URL, username, and password (stored via DPAPI). TimerStudio polls for events at the interval set by CalDavPollMinutes (default 15).

10.2 ICS URL Feeds

Add public ICS calendar URLs (Google Calendar, Outlook.com, etc.) in Settings → Calendar → ICS Feeds. Each URL is fetched and refreshed at CalDavPollMinutes.

10.3 Calendar Display Settings

Setting	Default	Description
LookaheadDays	7	Events within N days appear in the calendar view
ExcludeKeywords	(empty)	Comma-separated keywords; matching events are hidden
HideBirthdays	false	Suppress birthday events from CalDAV/ICS
HideHolidays	false	Suppress holiday events

11. Email Alerts

Mode	Code	Description
Disabled	0	No email alerts
SMTP	1	Send via configured SMTP server; password encrypted with DPAPI
Outlook COM	2	Use the installed Outlook desktop app's COM API
OAuth2	3	Authenticate via OAuth2 (Google/Microsoft); no password stored

11.1 SMTP Setup

Settings → Alerts → Email → SMTP: enter Host, Port, Username, Password (stored via Windows DPAPI), and toggle SSL/TLS. Test with the Send Test button.

11.2 OAuth2 Setup

Settings → Alerts → Email → OAuth2: click Authorise and complete the browser consent flow. Refresh tokens are stored in the Windows Credential Manager.

- Per-timer email recipient can differ from the global SMTP From address.

12. Webhooks

Set WebhookUrl on any timer. When the timer fires an alert, TimerStudio sends an HTTP POST with a JSON body:

```
{"timer":"Label","mode":"Countdown","phase":"Alert","timestamp":"2026-05-13T14:30:00Z","elapsed":3600
}
```

Webhooks can trigger Zapier/Make automations, Discord bots, Slack incoming webhooks, or any HTTP endpoint. No authentication is built in — use a secret in the URL if needed.

13. Overlay Windows

Overlay Type	Description
Lens Overlay	Per-alert floating card; dismisses automatically on interaction

Fullscreen Overlay	Full-screen dim with centred alert card; EscalateAfterSeconds triggers this
Timer List Overlay	Always-visible panel showing all timers with ShowInOverlay=true; useful for streaming

The Timer List Overlay is a borderless, always-on-top, click-through panel. Position it anywhere on screen. It updates live as timers count down. Toggle via hotkey Alt+O (configurable).

14. Statistics Dashboard

Open NavRow → Stats for a historical view of timer activity. The dashboard shows:

• Total focused time per category (Pomodoro work phases) • Completion rate (timers completed vs dismissed) • Daily/weekly heatmap of timer activity • Longest streak (consecutive Pomodoro work sessions)

Statistics are stored locally in %APPDATA%\TimerStudio\stats.db (SQLite).

15. Categories & Grouping

Assign each timer a Category (free-text label). Enable GroupedView in Settings to group timer cards by category in the main list. Collapse/expand categories using the arrow on the category header. Collapsed category IDs are remembered in CollapsedCategories.

Bulk operations: right-click a category header to Start All, Pause All, or Reset All timers in that category.

16. Hotkeys

Hotkey	Action
Ctrl+N	Add new timer
Alt+S	Open Settings
Alt+T	Cycle theme
Alt+H	Hide to tray
Alt+A	Open About dialog
Ctrl+Shift+1–5	Load preset 1–5
Space (card focused)	Start / Pause selected timer
Delete (card focused)	Delete selected timer (with confirmation)
Ctrl+D (card focused)	Duplicate selected timer
Ctrl+Z / Ctrl+Y	Undo / Redo timer configuration changes

17. Settings Reference

Setting	Type	Default	Description

UiScale	Enum	1.0	1.0/1.25/1.5/2.0
ThemeName	String	Dark	One of 20 themes
ThemeAuto	Bool	false	Follow Windows light/dark mode
GroupedView	Bool	false	Group cards by Category
CalDavUrl	String		CalDAV server URL
CalDavPollMinutes	Int	15	Calendar refresh interval
LookaheadDays	Int	7	Days ahead to show calendar events
HideBirthdays	Bool	false	Filter birthday events
HideHolidays	Bool	false	Filter holiday events
EmailMode	Enum	0	0=Off/1=SMTP/2=Outlook/3=OAuth2
SmtpHost	String		SMTP server hostname
SmtpPort	Int	587	SMTP port (587=STARTTLS, 465=SSL)
SmtpUsername	String		SMTP login username
SmtpPassword	String	(DPAPI)	DPAPI-encrypted SMTP password
LogLevel	Enum	Warning	Debug/Info/Warning/Error

18. Themes

All 20 built-in themes available. Timer card tints (CardTint) are additive overlays on top of the active theme's card background, so they work consistently across all themes.

19. Accessibility

Criterion	Level	Status
1.4.6 Contrast	AAA	All UI text ≥7:1 on all 20 themes
2.5.5 Target Size	AA	All controls ≥44×44 px; card buttons 44 px minimum
2.4.7 Focus Visible	AA	2 px Gold/Cyan focus ring on keyboard navigation
1.4.1 Use of Colour	A	Alert states use icon + text; CardTint has accessible label
2.1.1 Keyboard	A	All timer operations accessible without mouse
1.3.1 Info & Relationships	A	GroupBox labels, radio groups correctly marked for AT

20. Troubleshooting

Symptom	Fix
Timer alert doesn't fire	Check AlertSound != Silent; check system volume; check AlertRepeatLimit > 0
SMTP email not sending	Verify host/port/SSL settings; check firewall; use Send Test button
OAuth2 login fails	Clear tokens in Credential Manager; re-authorise in Settings → Email → OAuth2
CalDAV events not appearing	Verify URL, credentials; check CalDavPollMinutes; click Refresh Now
Webhook POST not received	Verify URL is HTTPS and accessible from this machine; check firewall
Fullscreen overlay covers apps	Set EscalateAfterSeconds to 0 to disable escalation; use Lens only

21. Log Files

Logs: %APPDATA%\TimerStudio\logs\timerstudio_YYYY-MM-DD.log

Statistics database: %APPDATA%\TimerStudio\stats.db

```
Log path: C:\Users\\AppData\Roaming\TimerStudio\logs\
```

22. Build Notes

Version v2026.05.13.0002 built with .NET 10, C# 14, WPF/MVVM. SMTP password encrypted via System.Security.Cryptography.ProtectedData (DPAPI). JSON source generation for settings and timer serialisation. OAuth2 tokens stored in Windows Credential Manager. Versioning: yyyy.MM.dd.NNN.

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